

IT Infrastructure & System Administration Internship

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MASTER OF COMPUTER APPLICATIONS



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BONAFIDE CERTIFICATE

This Internship report titled IT Infrastructure & System Administration by Utkarsh Srivastava [EA2432251010355] who carried out the Internship Work under my supervision along with the company mentor. Certified further, that to the best of my knowledge, the work reported herein does not form any other internship report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate

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Utkarsh Srivastava

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Abstract

This internship at **Ansys Software Pvt Ltd** provided valuable hands-on experience in the field of **Cloud Infrastructure Support using AWS and Azure**. The internship was aimed at enhancing my practical understanding of cloud computing concepts, system administration practices, and enterprise IT infrastructure management.

During the internship, I worked on provisioning and configuring **virtual machines**, managing **cloud storage solutions**, designing **virtual networks**, and setting up **identity and access management policies**. I was also involved in **monitoring system performance**, ensuring **data security and compliance**, and supporting **backup and disaster recovery operations**.

Through active collaboration with the IT Infrastructure team, I gained exposure to hybrid cloud environments, automation tools, and best practices in maintaining high availability and scalability. This internship not only strengthened my knowledge of AWS and Azure but also improved my problem-solving and technical support skills, which are essential for a career in **Cloud Infrastructure and System Administration**.

Introduction

Ansys Software Pvt Ltd is a global leader in engineering simulation software, delivering innovative solutions to help customers design .

and optimize products. The IT Infrastructure & System Administration team plays a critical role in enabling secure and scalable cloud services to

support global operations. My role as a System Administrator Intern included contributing to the setup, monitoring, and support of AWS & Azure resources.

This internship provided me with exposure to real-world enterprise cloud operations, including hybrid deployments and security management.

Analysis & Requirement

The Cloud Infrastructure Support internship required both technical and analytical skills for managing IT infrastructure in the cloud.

Skills & Knowledge Required:

- Networking fundamentals, Linux/Windows administration, and virtualization.
- AWS services such as EC2, S3, IAM, and VPC.
- Azure services such as Virtual Machines, Blob Storage, Virtual Networks, and Azure Active Directory.

Tools & Technologies Used:

- AWS Console, Azure Portal, CLI, and PowerShell.
- Cloud monitoring tools like AWS CloudWatch and Azure Monitor.

Requirements:

- High availability, secure authentication, scalability, disaster recovery, and cost efficiency were critical to the project.

Problem Description / Modules

The Cloud Infrastructure Support internship was divided into several modules to focus on specific aspects of cloud administration:

1. Identity & Access Management – Configuring AWS IAM and Azure Active Directory for secure authentication and role-based access.
2. Compute Management – Provisioning and maintaining AWS EC2 and Azure Virtual Machines.
3. Storage & Backup – Configuring AWS S3, Azure Blob Storage, and managing snapshots and backups.
4. Networking – Setting up VPCs, subnets, VPNs, security groups, and Azure Network Security Groups.
5. Monitoring & Logging – Implementing AWS CloudWatch and Azure Monitor for alerts and performance tracking.
6. Security & Compliance – Ensuring data encryption, patching, and compliance with company security standards.

Design Phase

In the design phase of the Cloud Infrastructure Support internship, the architecture and components of the cloud systems were defined. The design included:

1. Cloud Infrastructure Layout:

- AWS and Azure services were organized into Identity, Compute, Storage, and Networking modules.
- Logical and physical design diagrams were prepared using StarUML and AWS/Azure architecture icons.

2. Security Architecture:

- IAM roles and Azure AD groups were designed for secure access management.
- Policies were enforced for least privilege and MFA-based authentication.

3. Data Flow Design:

- Defined data transfer between S3, Blob, EC2, and Virtual Machines.
- Backup flow to ensure data redundancy and disaster recovery.

4. Monitoring Design:

- Dashboards and alarms were designed using AWS CloudWatch and Azure Monitor.

Implementation Phase

In this phase, the designed infrastructure was implemented using both AWS and Azure platforms.

1. AWS Implementation:

- Created IAM users, groups, and policies.
- Launched EC2 instances and configured S3 buckets.
- Set up CloudWatch for performance monitoring.

2. Azure Implementation:

- Configured Azure Active Directory for authentication.
- Created Virtual Machines and Azure Blob Storage containers.
- Implemented Azure Monitor for real-time metrics and alerts.

3. Automation:

- Used AWS CLI and Azure PowerShell scripts for automated provisioning.
- Configured scheduled backups using Lambda functions and Azure Automation.

Testing Phase

Testing was performed to ensure that all configured services and automation worked correctly.

1. Functional Testing:

- Verified IAM role permissions and access restrictions.
- Checked EC2/VM performance and connectivity.
- Tested backup and restore processes for data consistency.

2. Security Testing:

- Conducted penetration testing for open ports and unencrypted connections.
- Validated encryption in transit (TLS) and at rest (SSE, Azure Storage Encryption).

3. Load Testing:

- Used CloudWatch metrics and Azure Load Testing to check system performance under peak conditions.

4. Results:

- All modules passed security and performance benchmarks with minimal latency.

Tools and Technologies Used

During the internship, a range of tools and technologies were utilized for implementation and testing:

1. Cloud Platforms:

- Amazon Web Services (AWS)
- Microsoft Azure

2. Infrastructure Management:

- AWS Management Console
- Azure Portal
- AWS CLI
- Azure PowerShell

3. Monitoring & Logging:

- AWS CloudWatch
- Azure Monitor

4. Security & Compliance:

- AWS IAM
- Azure Active Directory

5. Design & Documentation Tools:

- StarUML (for diagrams)
- Microsoft Word and Excel (for reports)

Conclusion

The internship at **Ansys Software Pvt. Ltd.** offered a valuable opportunity to gain practical exposure to real-time **Cloud Infrastructure Support** using both **AWS** and **Microsoft Azure** platforms. Throughout the internship, I developed strong capabilities in cloud provisioning, identity and access management, system monitoring, data protection, and infrastructure automation.

This experience enabled me to understand how enterprise-level IT environments operate, including hybrid cloud management, security governance, backup strategies, and performance optimization. Working closely with the IT Infrastructure Team strengthened my problem-solving abilities, improved my technical confidence, and provided a deeper understanding of cloud operations and system administration best practices.

Overall, this internship was instrumental in enhancing both my technical knowledge and professional readiness. The hands-on tasks, real-time troubleshooting, and exposure to enterprise tools have helped me build a strong foundation for a future career in **Cloud Infrastructure, DevOps, or Senior System Admin.**

Appendices

Appendix A – Tools and Technologies Used

- **Cloud Platforms:** AWS, Microsoft Azure
 - **Compute Services:** EC2, Azure Virtual Machines
 - **Storage Services:** S3, Azure Blob Storage
 - **Identity Management:** IAM, Azure Active Directory
 - **Monitoring Tools:** AWS CloudWatch, Azure Monitor
 - **Automation Tools:** AWS CLI, Azure PowerShell
 - **Networking:** VPC, Subnets, NSGs, Route Tables
 - **Backup & DR Tools:** AWS Backup, Azure Recovery Services Vault
-

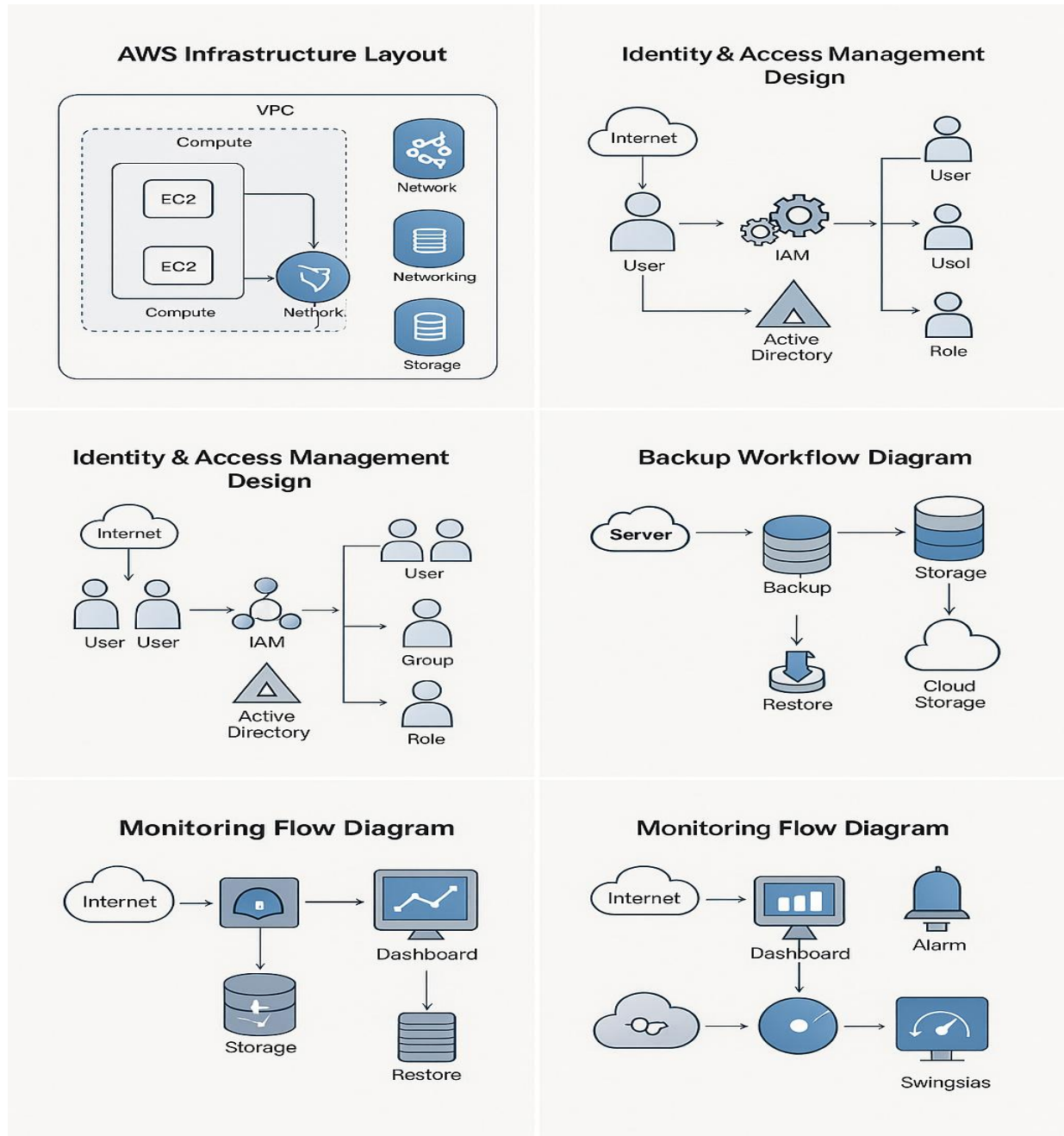
Appendix B – AWS Tasks Performed

- Created and managed IAM users, groups, roles, and policies
 - Provisioned EC2 instances and configured security groups
 - Designed and configured S3 buckets with lifecycle rules
 - Set up monitoring dashboards using CloudWatch
 - Automated provisioning using AWS CLI
 - Configured Lambda-based scheduled backup jobs
-

Appendix C – Azure Tasks Performed

- Created Azure AD users and assigned group-based access
 - Launched and configured Azure Virtual Machines
 - Created Blob Storage containers with access tiers
 - Configured NSGs and custom inbound/outbound rules
 - Implemented monitoring and alerts using Azure Monitor
 - Automated provisioning via Azure PowerShell scripts
-

Appendix D – Architecture Diagrams

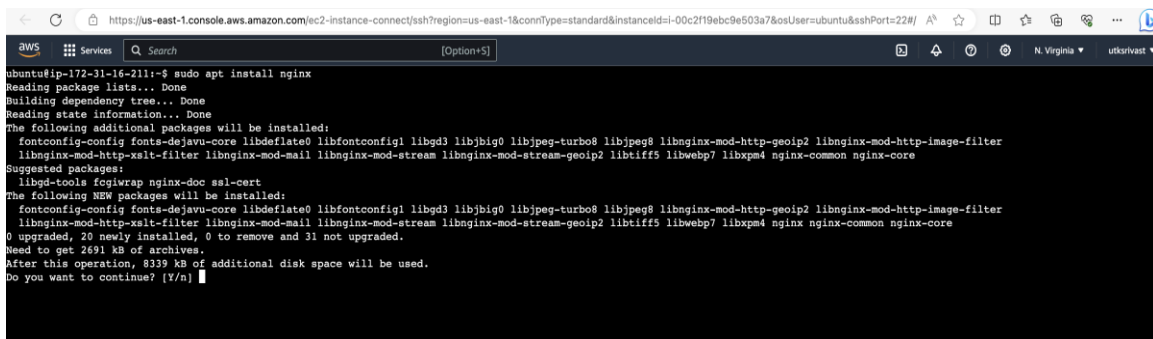


- AWS Infrastructure Layout
- Azure Infrastructure Layout
- Identity & Access Management Design
- Backup Workflow Diagram
- Monitoring Flow Diagram

Appendix E – Scripts & Configuration Samples

- AWS CLI Commands
- Azure PowerShell Commands
- IAM Policy JSON Templates
- Backup Automation Snippets

AWS CLI Interface –



```

https://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh?region=us-east-1&connType=standard&instanceId=i-00c2f19ebc9e503a7&osUser=ubuntu&sshPort=22#
AWS Services Search [Option+S]
ubuntu@ip-172-31-16-211:~$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fontconfig-config fonts-dejavu-core libdeflate0 libfontconfig1 libgd3 libjpeg8 libjpeg-turbo8 libjpeg9 libnginx-mod-http-geoip2 libnginx-mod-http-image-filter
  libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libtiff5 libwebp7 libxpm4 nginx-common nginx-core
Suggested packages:
  libgd-tools fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  fontconfig-config fonts-dejavu-core libdeflate0 libfontconfig1 libgd3 libjpeg8 libjpeg-turbo8 libjpeg9 libnginx-mod-http-geoip2 libnginx-mod-http-image-filter
  libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libtiff5 libwebp7 libxpm4 nginx nginx-common nginx-core
0 upgraded, 20 newly installed, 0 to remove and 31 not upgraded.
Need to get 2691 kB of archives.
After this operation, 8339 kB of additional disk space will be used.
Do you want to continue? [Y/n]
  
```

Appendix F – Screenshots

- AWS Management Console
- Azure Portal
- EC2 / VM Configuration
- S3 / Blob Storage Setup
- CloudWatch / Azure Monitor dashboards

EC2

The screenshot shows the AWS Management Console for the us-east-1 region. The left sidebar contains navigation links for EC2 Dashboard, Global View, Events, Instances, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a list of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, and Public IP. One instance, 'Web Server' (i-00c2f19ebc9e503a7), is shown in a 'Running' state. Below the list, the details for this instance are expanded, showing its configuration.

Instance: i-00c2f19ebc9e503a7 (Web Server)

Details | Security | Networking | Storage | Status checks | Monitoring | Tags

Instance summary Info

Instance ID i-00c2f19ebc9e503a7 (Web Server)	Public IPv4 address 54.89.9.24 open address	Private IPv4 addresses 172.31.16.211
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-54-89-9-24.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-16-211.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-16-211.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations.
Auto-assigned IP address 54.89.9.24 (Public IP)	VPC ID vpc-0dc0ae0a666f51f03	

The screenshot shows the 'Connect to instance' page in the AWS Management Console. The page title is 'Connect to instance' and it provides instructions on how to connect to the instance i-00c2f19ebc9e503a7 (Web Server). The 'EC2 Instance Connect' tab is selected, showing two connection options: 'Connect using EC2 Instance Connect' (selected) and 'Connect using EC2 Instance Connect Endpoint'. The 'Connect using EC2 Instance Connect' option is highlighted with a blue border. Below the connection options, the public IP address (54.89.9.24) and the user name (ubuntu) are displayed. A note at the bottom states: 'Note: In most cases, the default user name, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.' The 'Connect' button is visible at the bottom right.

Connect to instance Info

Connect to your instance i-00c2f19ebc9e503a7 (Web Server) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-00c2f19ebc9e503a7 (Web Server)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address
54.89.9.24

User name
Enter the user name defined in the AMI used to launch the instance. If you didn't define a custom user name, use the default user name, ubuntu.
ubuntu

Note: In most cases, the default user name, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel **Connect**

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